

So basically, my component does the prediction and early identification of landslides, so I can predict the early landslide using different areas or facts. There is rainfall, soil measures in different areas, and the kind of rainfall that triggers a landslide. When I try to understand my component, I'm going to the feasibility study for that. So, the feasibility study is divided into 5- 6 phases. There are 1. facts or the objective of my component, 2. Facts and reasons why a landslide occurs, and for which areas 3. Technology that I'm going to use 4. Dataset or data resources 5. subcomponents

1. Objective of my component

Our whole Research is based on the Integrated Disaster Management and Emergency Response System. So when we come to that topic, my component is more effective on the Integrated Disaster Management Part. So my component objective is to identify the earlier landslide and reduce the damage to society or the land. So then we need to identify the landslide first, so we should consider the different areas. Types of landslides, which areas are more prone to landslides, how much rainfall is needed to trigger landslides, what kinds of soil can cause landslides quickly, and what human errors or mistakes can occur to help the landslide. Etc.

2. Facts That Cause Landslides

When we come to this point, I'm going to discuss some areas. Types of landslides can occur.
Falls

- With vertical or steep slopes such as cliffs, a landslide will often take the form of a fall – this is where material collapses from the slope, falls downwards, and collects towards the base. As a result, we can recognise the occurrence of a fall and differentiate it from the other landslide types by the accumulation of earth, rock, and debris at the bottom of a steep slope.

Topples

- As the name suggests, a topple is a type of landslide where the rock, earth, or debris rotates forward away from the slope and literally topples downwards. Under normal circumstances, there will be an axis at or close to the bottom of the slope about which the rotation occurs.

Next Landslide Calls Slides That Slide are divided into two categories: rational or Translational.

Rational

- We can spot rotational slides in cases where the slip surface is curved (often like the surface of a spoon) – this is referred to as a listric rupture or slip surface. As a result of this appearance, rotational landslides are sometimes called ‘slumps’.

Translational

- Where a landslide occurs along the surface of a fault, joint, or bedding plane, it is said to be translational or ‘planar’. Often causing huge damage, translational landslides can result in the rapid movement of material down a slope. The speed of the slide ultimately depends on how deep the underlying fault is.

Link For Resources - <https://www.tensarinternational.com/resources/articles/what-is-a-landslide>

When it comes to Sri Lanka, the most common types of landslides are mostly common types of landslide that occur, and rotational and translational types are common. National Building Research Organization (NBRO) Research proves that.

According to most articles and researchers, Badulla, Nuwareliya, Rathnapura, and Kegalle districts are the most affected by that disaster, most of the time in the past 20 years.

Badulla Districts

- Koslanda & Meeriyabedda: Site of a massive landslide that destroyed entire plantation settlements.
- Passara & Lunugala: Mountainous towns are highly prone to major failures blocking main transportation routes.
- Haldummulla & Ella: Sizable tourist and transit towns frequently hit by earth slips and boulder falls during monsoon surges

Kegalle District

- Aranayaka: Infamous for the catastrophic 2016 Samasara Hill landslide that completely buried several village areas.
- Mawanella & Bulathkohupitiya: Areas that face severe cutting failures and flash mudflows stemming from loose, rubber-plantation soils

Nuwara-Eliya District

- Walapane & Hanguranketha: Frequently experience massive slope collapses, structural cracks in homes, and road blocks.
- Kotmale: Heavily monitored due to the risk of unstable hillsides abutting the large Kotmale reservoir.

- Talawakele & Hatton: Tea-growing towns that encounter localized landslides and soil erosion annually

Rathnapura District

- Balangoda & Imbulpe: Mountainous terrains framing the transition to the hill country, experiencing frequent translational soil slides.
- Eheliyagoda & Kuruwita: Gem-mining and heavy-rainfall hubs where the ground collapses easily when saturated by water.

What amount of rainfall causes landslides

- 75 mm within 24 hours (Level 1 - Watch): An alert is triggered if continuous rain reaches this level, indicating the soil is becoming heavily saturated.
- 100 mm within 24 hours (Level 2 - Alert): Slopes become highly unstable, and residents are told to stay vigilant for signs of earth slips.
- 150 mm within 24 hours OR 75 mm within just 1 hour (Level 3 - Evacuation): This extreme volume triggers an immediate evacuation order because a catastrophic landslide can occur at any moment.

The Heavy Rain Fall Is the Most Accurate or the powerful reason for those landslides, but other reasons have helped to cause them.

Human Errors

- Dangerous Slope Cutting: Excavating the base of hills to build homes or widen roads destroys natural structural support, leading to widespread "cutting failures".
- Deforestation: Stripping mountain forests for commercial crops or housing removes root systems that act as anchors holding the topsoil together.
- Unplanned Drainage Blocks: Building houses close together without coordinated gutters causes wastewater to pond and sink directly into the hillside, dangerously increasing internal soil weight.
- Improper Land Use & Zoning: Expanding settlements on landslide-prone mountain terrains without structural clearance or engineering safety guidelines from the [NBRO](#)

In the Past 20 years, the most devastating Single landslide event in Sri Lanka was the Samasarakanda landslide in Aranayaka in Kegalle District in 2016.

127 Human Lives have been taken away because of that disaster

Three entire villages (Siripura, Pallegage, and Elangapitiya) were completely buried

75 houses were destroyed, and over 500 families were forcibly displaced

Key Reasons For the Pandemic

- **Extreme Rainfall:** A slow-moving tropical depression dumped a massive 446.5 mm of rain over just three days leading up to the failure.
- **Deep Geological Weathering:** Scientific soil testing later proved that the mountain's bedrock (granitic gneiss) had suffered intense historical tectonic crushing. This left a heavily fragmented, deeply weathered subterranean layer that completely gave way under water pressure.
- **Massive Travel Distance:** Because the soil turned into a liquid slurry (debris flow), the landslide mass traveled an unusually long 2-kilometer distance down the mountain, crushing villages located far beneath what was considered the immediate hazard zone

I asked a question from Google AI mode about whether this pandemic could have been identified earlier.

This is the Answer

- **Lack of Micro-Level Forecasting:** While the National Building Research Organisation (NBRO) had issued broad, district-level regional landslide warnings due to the heavy rain, there was no technology in place to identify that *Samasarakanda Mountain* specifically was reaching its breaking point. [
- **False Sense of Security:** The local communities had lived on those slopes for generations without experiencing a major landslide, leading them to believe the mountain was safe.
- **The Velocity of Failure:** The landslide initiated at the top of the mountain and transformed rapidly into a lightning-fast debris flow. Once the slope cracked, the mass traveled the 2 km distance so quickly that families down the valley had zero time to run.

So, that is the first Point That tells of the lack of **Micro-Level Forecasting**.

This is our point. By doing our research using the technology, we can forecast and predict it. So, from that, we can reduce the Pandemic or the disaster. We Can Save Innocent Lives. So, from that, we are going to research how we can Do Integrated Disaster Management Using Modern Technologies.

Next Main Part Is

DataSet Collectings And DataResources

<https://liris.nbro.lk/en> - nbro Portal

If we need the Accurate data, we should send a letter in a formal way through Email and slide point locations, hazard zones, and rainfall data about those dataset requests.

Primary Contacts for Data Requests

- Director General (NBRO)
- Email: dgnbro@slt.lk
- Phone: +94 11 250 5149
- Director of LRRMD (Landslide Research & Risk Management Division)
- Email: info@nbro.gov.lk (Note: Mark the subject line clearly: "Attn: Director, LRRMD - Landslide Dataset Request")
- Phone: +94 11 258 8946 (Extension: 666)

Department of Meteorology

- Landslides in Sri Lanka are primarily rainfall-triggered. [[1](#)]
- **What you can get:** High-resolution historical weather records, daily rainfall numbers, and spatial rainfall maps.
- **How to access:** You can submit a public data request to the [Department of Meteorology](#) to cross-reference weather anomalies with known landslide timelines.

Tools And Technology We Should Use to Implement Our Research

Building a predictive or forecasting model involves a combination of Geographic Information Systems (GIS) and Machine Learning (ML)

Forecasting = GIS + ML

GIS software (Geographic Information System) is a digital tool used to create, manage, analyze, and map all types of location-based data

It helps to layer data -

GIS works like a stack of transparent maps. Each layer represents a different type of information about the same geographic location.

For my landslide project, I would stack a digital elevation layer (slopes) on top of a geology layer (rock types), on top of a weather layer (rainfall maps), on top of a satellite photo.

It helps to give an accurate answer to the asked question widely -

Normal Database doesn't give the most accurate answer because of the limitations of wide Ranges

As IT undergraduates, the most accurate GIS is

Python + PostGIS

For Full-Stack Web Development: MapLibre GL JS / Leaflet

Why it fits IT: MapLibre GL JS uses WebGL to render massive datasets smoothly in 3D using vector tiles, which is critical for showing terrain slopes accurately. Leaflet is perfect if you need a lightweight, fast, and highly customizable JavaScript map.

Oh! I have missed a Point

Soil Types That Turn Landslides Swiftly

Research Gate -

https://www.researchgate.net/publication/366114343_Geoinformatics_approach_to_classifying_landslide_hazard_in_The_Hantana_Mountain_and_its_vicinity_in_Sri_Lanka

Red-Yellow Podzolic Soils (Districts: Badulla, Nuwara Eliya, Kandy): This is the most critical soil classification for Sri Lankan landslide models. It covers the steeply dissected, hilly, and rolling terrains of the Central Highlands. This soil features a heavy accumulation of leached clay minerals in its sub-layers. When monsoonal rains hit, these clay sub-layers trap water, build immense pore-water pressure, and turn into a slippery slide surface.

Mountain Regosols (Districts: Kegalle, Ratnapura): Found on steep, highly mountainous slopes. These are immature, shallow soils mixed with gravelly rock fragments (colluvium). They have weak structural binding, meaning they lose cohesion and liquefy rapidly into debris flows when saturated.

Another point is the slope angle in degrees.

There is a table showing the angle that indicates the risk levels released by the NBRO.

So from that, I am going to do my component based on different inputs and parameters like soil type, type of landslide, the angle of a slope, targeted areas, and so on.

Slope Angle Range 	NBRO Risk Level	Machine Learning Weight	Physical Process
> 35°	Very High Risk	10 / 10	Steep escarpments, extreme gravity, highly prone to fast debris flows.
26° – 35°	High Risk	8 / 10	The optimal angle where thick soil layers rest on bedrock and easily fail under heavy rain.
17° – 25°	Medium Risk	5 / 10	Moderately safe unless disturbed by human slope cutting.
11° – 16°	Low Risk	2 / 10	Gentle slopes, rarely slide unless completely deforested.

So, finally, my objective is to Turn This Physical Geomatics system into an accurate and Powerful Predictive and Forecasting Model.